Discussion 1

MSBA 402

Welcome

What discussion can covers:

- Lecture recap
- Practice problems
- Homework support
- Technical support

Also utilize Piazza and office hours. Bruin Learn access?

Email fauzan.prasetya@ucla.edu

TA office hours on Thursday 11am to 12pm

Today’s discussion overview

This discussion will cover these topics:

- Intro to databases
- Intro to relational models
- Keys

Database overview

Can do:

- Create
- Read
- Update
- Delete

Provide one single location → one database (DB) can consist of multiple tables

Schema: overall design of the database.

Some people will refer schema as for table → Table schema

Data modelling

More detail about how we design the database.

Some examples:

- Define **data** → defining table name, what are the columns and rows represent
- Encode **relationship** → using a key in one table to reference detailed information stored in another table
- Impose **constraints** → one column can only contain numbers, can one column contains missing values

Customers

CustomerID	FirstName	LastName	StreetAddress	City	State	ZipCode
1010	Angel	Kennedy	667 Red River Road	Austin	TX	78710
1011	Alaina	Hallmark	Route 2, Box 203B	Woodinville	WA	98072
1012	Liz	Keyser	13920 S.E. 40th Street	Bellevue	WA	98006
1013	Rachel	Patterson	2114 Longview Lane	San Diego	CA	92199
1014	Sam	Abolrous	611 Alpine Drive	Palm Springs	CA	92263
1015	Darren	Gehring	2601 Seaview Lane	Chico	CA	95926

ROWS

COLUMNS

KEYS

Customers

CustomerID	CustFirstName	CustLastName	CustPhone	<<other columns>>
10001	Doris	Hartwig	555-2671	...
10002	Deb	Waldal	555-2496	...
10003	Peter	Brehm	555-2501	...
<< more rows here >>				

RELATIONSHIP

Engagements

EngagementNumber	CustomerID	StartDate	EndDate	StartTime	<<other columns>>
3	10001	2016-09-10	2016-09-15	13:00	...
13	10003	2016-09-17	2016-09-20	20:00	...
14	10001	2016-09-24	2016-09-29	16:00	...
17	10002	2016-09-29	2016-10-02	18:00	...
<< more rows here >>					

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DB Schema example

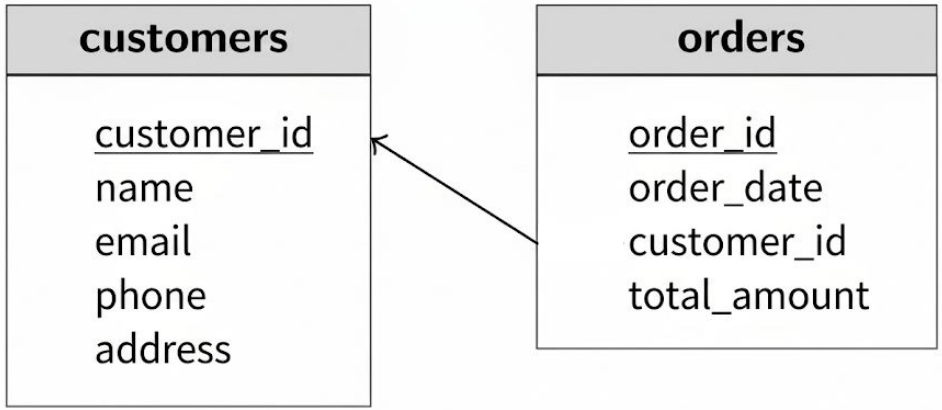


Table schema example

Column Name	Data Type	Nullable?	Key / Constraint	Description
order_id	INT	No	Primary Key	Unique numeric identifier for internal database tracking
order_number	VARCHAR(20)	No	Unique	Human-readable order reference code (e.g., ORD-2026-9481)
customer_id	VARCHAR(10)	No	Foreign Key	References customers.customer_id
order_date	DATETIME	No	Default: NOW()	Timestamp when the order was placed
total_amount	DECIMAL(10, 2)	No	Check (>= 0)	Total monetary value of the purchase
payment_method	VARCHAR(30)	Yes	—	Method used (e.g., Credit Card, PayPal, NULL if uncompleted)
order_status	VARCHAR(20)	No	Default: 'Pending'	Current status (e.g., Pending, Shipped, Delivered, Refunded)
shipping_country	VARCHAR(50)	No	—	Destination country for order delivery

Primary Key (PK)

Definition:

Single column (or a simplest set of columns) that **uniquely** identify each row in table

In other words:

1. Unique (no duplicates)
2. Cannot be null
3. Can be one column or multiple

If multiple → must be the simplest, or maintained by the institution



- Table Agents contains list of Agents

  → PK is AgentID

- Table Entertainers contains list of Entertainers

  → PK is EntertainerID

To find any PK in a table, use **context.**

What does each row represent?

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Primary Key (PK) example

What is the Primary Key of this table?

order_id	customer_id	order_date	total_amount	payment_method	order_status	shipping_country
1003	CUS-802	2026-03-02 14:20:11	210.00	Credit Card	Shipped	USA
1004	CUS-109	2026-03-02 18:05:40	18.25	Debit Card	Delivered	UK
1005	CUS-633	2026-03-03 08:30:00	520.00	Bank Transfer	Processing	Germany
1006	CUS-214	2026-03-03 11:12:19	89.99	Credit Card	Cancelled	USA
1009	CUS-109	2026-03-05 15:22:50	74.10	Debit Card	Refunded	UK
1010	CUS-554	2026-03-06 19:10:02	310.50	Credit Card	Shipped	Australia

Primary Key (PK) example

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Foreign Keys (FK)

Purpose:

~~To merge one table into another~~ → This is incorrect.

Foreign keys are used to impose referential integrity constraints.

Referential integrity: a rule ensuring all relationships between tables in DB are valid.

Values of a column (or set of columns) in the referring table must exist in the referred table.

Referred table must exist before referring table.

Customers Referred table as parent

CustomerID	CustFirstName	CustLastName	CustPhone	<<other columns>>
10001	Doris	Hartwig	555-2671	...
10002	Deb	Waldal	555-2496	...
10003	Peter	Brehm	555-2501	...
<< more rows here >>				

Engagements Referring table as child

EngagementNumber	CustomerID	StartDate	EndDate	StartTime	<<other columns>>
3	10001	2016-09-10	2016-09-15	13:00	...
13	10003	2016-09-17	2016-09-20	20:00	...
14	10001	2016-09-24	2016-09-29	16:00	...
17	10002	2016-09-29	2016-10-02	18:00	...
<< more rows here >>					

- If we change CustomerID 10001 in **Customers** table to random number like 11111,

  → what will happen?

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<< more rows here >>					

- If we change CustomerID 10001 in **Customers** table to random number like 11111,

  → engagement number 3 and 14 became **orphans** (without parents)

- If we change CustomerID 10001 in **Engagements** table to random number like 11111 ,

  → engagement number 3 and 14 may **no longer match** any to any customer

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Unique Key and Natural Key

Order summary

Order #: 112-8051483-6638635

Ordered September 27, 2026 1:16PM

Item(s) Subtotal:	$26.58
Delivery Fee:	~~$13.95~~ $0.00
Bottle Deposit Fee	$0.10
Total before tax:	$26.68
Estimated tax to be collected:	$0.00
Driver tip:	$3.00
Grand Total:	**$29.68**

Unique Key

Unique key must be **unique**.

They don’t serve as Primary Key.

Other example?

order_id	order_number	customer_id	order_date	total_amount
1003	ORD-2026-9481	CUS-802	2026-03-02 14:20:11	210.00
1004	ORD-2026-3105	CUS-109	2026-03-02 18:05:40	18.25
1005	ORD-2026-7240	CUS-633	2026-03-03 08:30:00	520.00
1006	ORD-2026-1183	CUS-214	2026-03-03 11:12:19	89.99
1009	ORD-2026-5592	CUS-109	2026-03-05 15:22:50	74.10
1010	ORD-2026-8047	CUS-554	2026-03-06 19:10:02	310.50

Natural Key

Natural key is used in **real-world business** contexts.

They frequently intuitive and recognizable enough for end-users.

Other example?

order_id	order_number	customer_id	order_date	total_amount
1003	ORD-2026-9481	CUS-802	2026-03-02 14:20:11	210.00
1004	ORD-2026-3105	CUS-109	2026-03-02 18:05:40	18.25
1005	ORD-2026-7240	CUS-633	2026-03-03 08:30:00	520.00
1006	ORD-2026-1183	CUS-214	2026-03-03 11:12:19	89.99
1009	ORD-2026-5592	CUS-109	2026-03-05 15:22:50	74.10
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Surrogate Key

Surrogate key is **automatically created** by the database.

It has no business meaning.

Frequently serve as Primary Key.

order_id	order_number	customer_id	order_date	total_amount
1003	ORD-2026-9481	CUS-802	2026-03-02 14:20:11	210.00
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1009	ORD-2026-5592	CUS-109	2026-03-05 15:22:50	74.10
1010	ORD-2026-8047	CUS-554	2026-03-06 19:10:02	310.50

Candidate Keys

Definition:

A **column** or **set of columns** that can **uniquely** identify every single row in a table.

One must serve as Primary Key.

If it set of columns, it should be **minimal**. Removing any column from the set destroys the uniqueness. No redundancy

Candidate Keys:

1. Employee_ID (PK)
2. Social_Security_Number
3. Email
4. {Office_Building, Desk_Number}

4 only works when no two employees can share the exact same desk in the same building.

Employee_ID	Social_Security_Number	Email	Full_Name	Department	Office_Building	Desk_Number
101	987-65-4321	alex@company.com	Alex Morgan	Engineering	Building A	12
102	876-54-3210	taylor@company.com	Taylor Swift	Marketing	Building A	15
103	765-43-2109	alex.m@company.com	Alex Morgan	HR	Building B	12

Candidate Keys example

What are the candidate keys?

order_id	order_number	customer_id	order_datetime	total_amount	payment_method	order_status	shipping_country
1003	ORD-2026-9481	CUS-802	2026-03-02 14:20:11	210.00	Credit Card	Shipped	USA
1004	ORD-2026-3105	CUS-109	2026-03-02 18:05:40	18.25	Debit Card	Delivered	UK
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1010	ORD-2026-8047	CUS-554	2026-03-06 19:10:02	310.50	Credit Card	Shipped	Australia

Candidate Keys example

What are the candidate keys?

1. Order_id
2. Order_number
3. {Customer_id, order_datetime}

3 only works when the system prohibits one customer making multiple orders in the same time

order_id	order_number	customer_id	order_datetime	total_amount	payment_method	order_status	shipping_country
1003	ORD-2026-9481	CUS-802	2026-03-02 14:20:11	210.00	Credit Card	Shipped	USA
1004	ORD-2026-3105	CUS-109	2026-03-02 18:05:40	18.25	Debit Card	Delivered	UK
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